

L13 - Linearity and Superposition

17 March 2024 15:52

Core Principles: Ideal circuits

Linearity

- Ratio between any two voltages or currents in a circuit is constant

$$\frac{V_1}{V_2} = c \quad V_1 = 5V \quad V_2 = 2V \quad \frac{V_1}{V_2} = \frac{5}{2} = 2,5$$

$$\frac{I_1}{I_2} = c \quad I_1 = 1,5A \quad I_2 = 0,6A \quad \frac{I_1}{I_2} = \frac{1,5}{0,6} = 2,5$$

Superposition

- Each independent source contributes to the voltage/current across an element
- The voltage/current is the sum of the contributions of each independent source

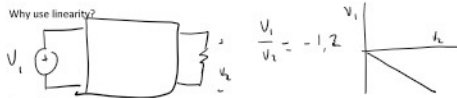
$$V_{out} = V_{out1} + V_{out2} + V_{out3} + \dots$$

Turn off voltage source - replace with short circuit

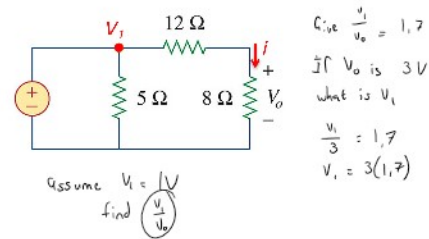
Turn off current source - replace with open circuit

Never turn off dependent sources

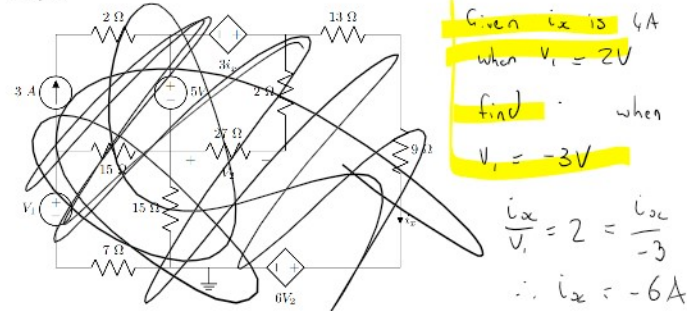
Why use linearity?



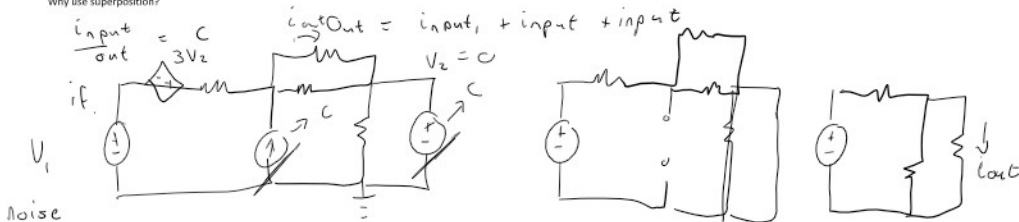
Example 1:



Example 2:

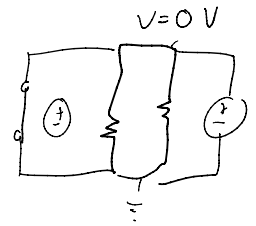
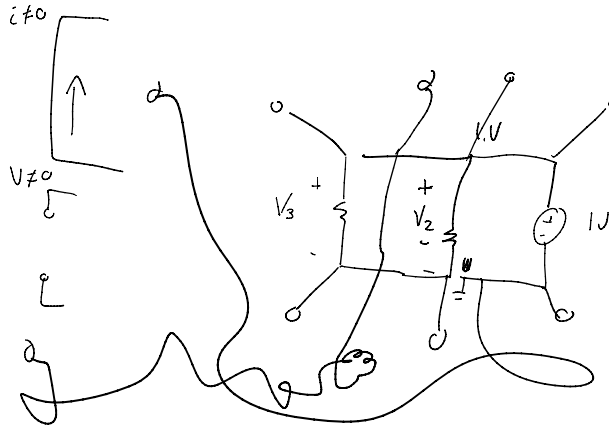
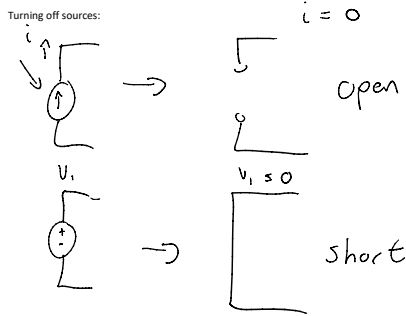


Why use superposition?

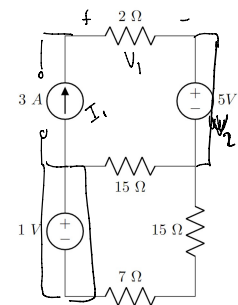


never turn it off

Turning off sources:

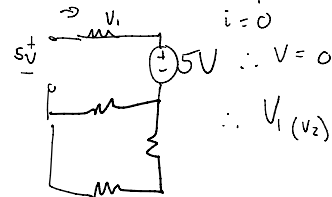


Example 1:



① find independent contribution

of 5V source to V_1 :



$$V_1(I_1) = 2V$$

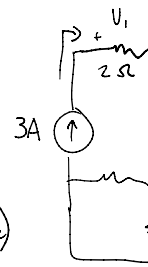
$$\therefore V_1 = V_1(I_1) + V_1(I_2) + V_1(V_2)$$

$$\therefore V_1 = 3.2 = 2 + 6 + 0 = 8V$$

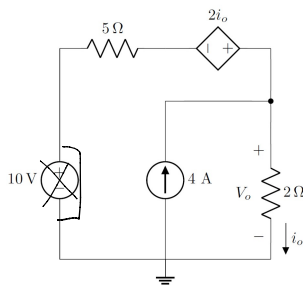
$$V_1(I_2) = 6V$$

② find contribution of I_1 (current source) to V_1 ?

$$V_1(I_2)$$

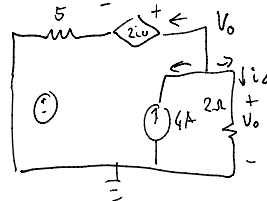


Example 2:



Find the contribution of 4A source

to i_o :



$$\frac{V_o - 2i_o}{5} + (-4) + \frac{V_o}{2} = 0$$

$$i_o = \frac{V_o}{2}$$

$$\frac{V_o - 2(\frac{V_o}{2})}{5} - 4 + \frac{V_o}{2} = 0$$

$$\frac{V_o}{2} = 4$$

$$i_o = 4A$$

$$V_o = 8$$

$$i_o(4A) = 4A$$